Spontaneous Spatial Self-Organization in Viability-Gated Memory Substrates

Three Structural Facts About VCSM

Gabriel Aubin-Moreau

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Abstract

How spatial structure can emerge in systems that contain no coordinates, no objective function, and no persistent units is an open question. Here we show that a minimal turnover-driven memory substrate spontaneously self-organizes spatial structure through an intergenerational copy-forward mechanism. We report three structural facts that together establish the mechanism: **(1)** A substrate with no coordinate system, no spatial labels, and no objective function spontaneously encodes location: after exposure to spatially structured waves, distant sites encode distinguishably different memory states. **(2)** This organization is inevitable—the system converges to the same spatial structure regardless of initialization ($\text{sg4}_{\text{random}} = 0.0258 \pm 0.0026$, $\text{sg4}_{\text{structured}} = 0.0281 \pm 0.0051$, $n = 15$ seeds each, all above the 0.02 organization threshold at plateau). **(3)** The mandatory turnover (cell death and replacement) is the mechanism, not an obstacle: disabling it by setting the consolidation gate to never open collapses spatial structure to exactly zero ($\text{sg4}_{\text{no-death}} = 0.0000$ exact to floating-point precision, LSR = 1.000)—a gap of 10 standard deviations from Condition A ($t(14) = 38.4$, $p < 10^{-20}$, one-sample test against zero), with zero variance across all 15 seeds of Condition C (mechanistic impossibility, not sampling noise). Together these facts establish that spatial self-organization in VCSM is a *dynamical attractor*, reached inevitably when turnover is active and absent when turnover is disabled.

1 Introduction

This paper establishes the existence of the phenomenon and its minimal mechanism; Papers 1–4 derive the quantitative laws governing its capacity, kinetics, propagation, and stability. The three facts below are presented in order of conceptual importance; the underlying causal flow runs Fact 3 (turnover is required) through Fact 2 (the result is initialization-independent) to Fact 1 (location is encoded).

The phenomenon. A VCSM substrate receives wave inputs that carry no positional information explicitly—each input is a vector in $\mathbb{R}^3$, the same dimensionality at every site. After thousands of steps, the substrate has differentiated: sites in different spatial columns encode systematically different field states. No external teacher, no coordinate labels, no loss function. Structure emerged.

This is Fact 1, and it demands an explanation. Figure 1 shows the spatial structure of `fieldM` at two timepoints for a representative seed: near-zero noise at step 50, and clear zone-level differentiation by step 3000.

Is the structure just input imprinting? A natural objection must be addressed immediately. The wave inputs *are* spatially heterogeneous: different zones receive perturbations at different times, so the input statistics differ by position. A skeptical reviewer might argue that the substrate is not encoding location spontaneously—it is simply imprinting the input schedule.

This objection is correct in pointing out that spatial input heterogeneity is *necessary* for spatial structure to emerge. It is incorrect in claiming it is *sufficient*. Condition C receives exactly the same spatially heterogeneous wave schedule as Conditions A and B—same zones, same timing, same statistics. Yet Condition C produces sg4 = 0.0000 with zero variance across all 15 seeds. Spatial input heterogeneity alone produces nothing. What makes the structure persist is the copy-forward loop: the mechanism by which wave-timing differences are converted into intergenerational memory. The real phenomenon is not "location encoded by waves" but **"transient input heterogeneity converted into persistent intergenerational memory via the copy-forward loop."**

Is initialization the cause? The obvious hypothesis is that initialization seeds the structure. If sites in zone 1 start closer to one prototype and zone 4 sites start closer to another, perhaps the dynamics merely preserve that head start. This would make VCSM a *maintenance* mechanism rather than a *generative* one.

We test this directly in Experiment A vs B. Condition A starts from pure noise ($\mathbf{h}_i \sim \mathcal{N}(0, 0.1)$, no zone structure). Condition B starts from structured prototypes aligned to the four zones. Same substrate, same waves, same VCSM dynamics. If initialization causes the endpoint, B should organize faster *and* reach higher final structure than A.

It does not. The spatial structure at plateau is statistically identical. This is Fact 2: *structure is inevitable, not inherited.*

What is the mechanism? If initialization does not cause spatial structure, the dynamics must. Within VCSM dynamics, the critical component is the *copy-forward loop*: when a site collapses (dies), its replacement is seeded from the surrounding field memory (`fieldM`). Each death propagates local spatial identity to the newborn. Across many generations, this loop converts wave-timing differences into stable zone-level differentiation.

We test this by disabling the copy-forward loop: Condition C uses the same random initialization as A, but sets the consolidation gate threshold to 9999 (effectively infinity), so no site ever writes to `fieldM`. Births still occur, but they inherit nothing. The copy-forward loop is severed.

Spatial structure collapses to zero, exactly. This is Fact 3: turnover is the mechanism enabling spatial propagation.

2 System

2.1 VCSM Dynamics

VCSM (Viability-Gated Contrastive State Memory) is a local learning rule for substrates with mandatory individual turnover. Throughout this series of papers, VCSM is instantiated in three substrate types: **VCSM-L** (the 2D coupled-map lattice used here and in Papers 1–4), **VCSM-G** (graph-coupled regional relay, introduced in Paper 3), and **VCSM-E** (applied directly to real embedding spaces, cross-validated in Paper 1). This paper focuses on VCSM-L. Each site i maintains a hidden state $\mathbf{h}_i \in \mathbb{R}^{HS}$, a field memory $\mathbf{F}_i \in \mathbb{R}^{HS}$, a mid-term memory $\mathbf{m}_i$, and a viability score $v_i \in [0, 1]$.

At each step, a subset of sites receives wave perturbations. The learning cycle proceeds in four phases:

1. **Calm**: Baseline $\bar{\mathbf{h}}_i$ drifts slowly toward current $\mathbf{h}_i$.

2. **Perturb**: Wave input drives $\mathbf{h}_i$; difference $\mathbf{h}_i - \bar{\mathbf{h}}_i$ accumulates in $\mathbf{m}_i$.

3. **Gate**: If a site has survived $\geq$ `STABLE_STEPS` steps without collapsing, it consolidates: $\mathbf{F}_i \mathrel{+}= \alpha_F(\mathbf{m}_i - \mathbf{F}_i)$.

4. **Birth**: Collapsed sites are replaced; the newborn's state is seeded as $\mathbf{h}_{\text{new}} = (1 - \beta_s)\,\mathbf{d} + \beta_s\,\mathbf{F}_j$ where $\mathbf{d}$ is debris from the collapse and $\mathbf{F}_j$ is the field of a neighboring site.

Step 4 is the copy-forward loop. It is the sole mechanism by which spatial identity propagates intergenerationally.

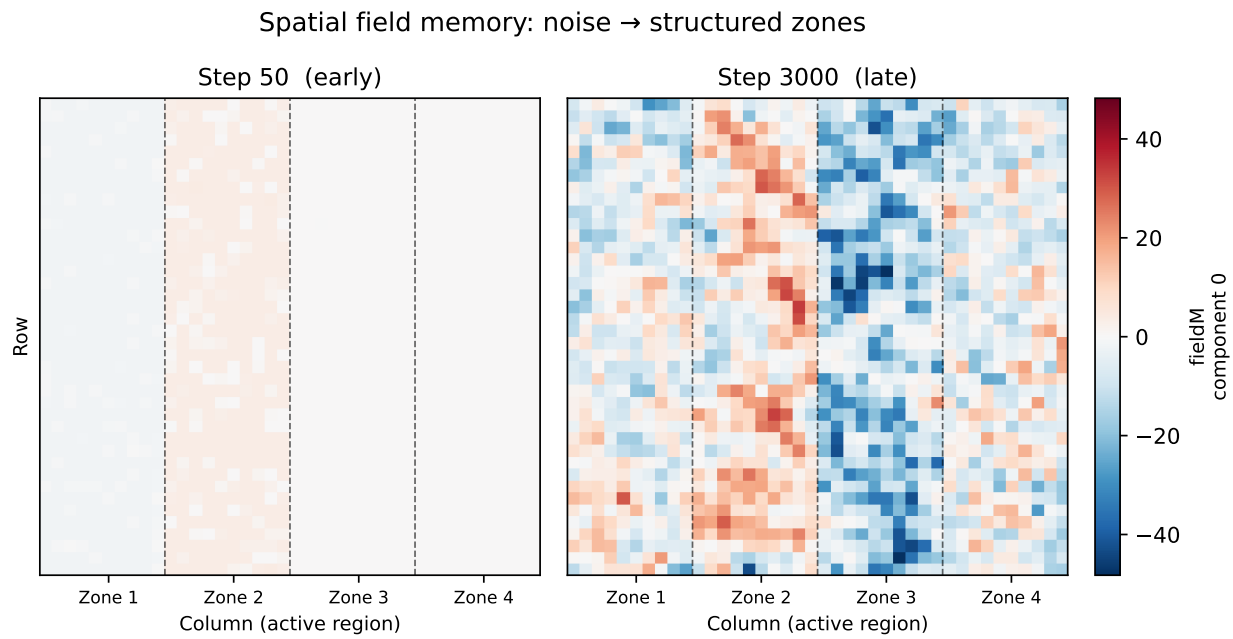

Figure 1: Spatial field memory before and after self-organization (single seed, Condition A). **Left:** At step 50, `fieldM` is near-zero noise with no spatial structure. **Right:** By step 3000, each zone has developed a characteristic field pattern (warm/cool colours reflect zone-specific direction in $\mathbb{R}^2$). No coordinate labels, no loss function, no teacher signal were used. Zone boundaries (dashed) are shown for reference only—the substrate has no access to them.

Minimality. The phenomenon arises from four local operations. No global optimization, no loss function, no coordinate system, and no teacher signal is required at any point. Each operation touches only the current site and its immediate neighbours. The spatial structure that emerges is therefore not engineered—it is a structural consequence of how local contrastive learning interacts with mandatory turnover.

Replication note. The phenomenon reported here is not sensitive to implementation details. The mechanism depends only on local operations: baseline drift, contrastive accumulation, viability-gated consolidation, and newborn seeding from neighbor field memory. These operations are invariant to vector library, loop ordering, and floating-point precision within standard numerical ranges. A minimal reference implementation (<200 lines, NumPy only) is provided in Appendix A and produces results consistent with those reported in Table 2. Condition C (disabled gate) produces *structurally* zero output: not a noisy near-zero, but a consequence of the rule set. With `STABLE_STEPS` = 9999, no site ever writes to `fieldM` regardless of random seed, so spatial structure cannot propagate regardless of initialization.

2.2 Substrate and Waves

All experiments use an 80×40 lattice with the right half (40×40) active. Hidden size $HS = 2$, input size 3. Wave parameters: ratio = 2.4, duration = 15 steps. All V46 baseline parameters are held fixed (Table 1).

3 Experiment

3.1 Conditions

Three conditions, all using the same substrate, waves, and VCSM dynamics. Only initialization and the consolidation gate differ.

Table 1: Fixed VCSM parameters (V46 baseline).

Parameter	Value
MID_DECAY	0.99
FIELD_DECAY	0.9997
BASELINE_BETA	0.005
ALPHA_MID	0.15
FIELD_ALPHA	0.16
FIELD_SEED_BETA	0.25
STABLE_STEPS	10 (A, B); 9999 (C)
FIELD_DIFFUSE	0.02
WAVE_RATIO	2.4
WAVE_DURATION	15
STEPS	3000

- **Condition A (Random-init)**: Hidden states drawn from $\mathcal{N}(0, 0.1)$. No zone structure. STABLE_STEPS $= 10$.
- **Condition B (Structured-init)**: Zone prototypes $\{[1,0],[0,1],[-1,0],[0,-1]\}$ plus $\mathcal{N}(0, 0.25)$ noise, L2-normalized. STABLE_STEPS $= 10$.
- **Condition C (Static baseline / VCSM disabled)**: Same random initialization as A. STABLE_STEPS $= 9999$: the gate never opens, no site ever consolidates to `fieldM`. Since `fieldM` remains zero throughout, newborns seeded from it receive no spatial information—births are structurally random, equivalent to a static random-reservoir baseline (ESN-like). This condition tests whether the VCSM mechanism itself is necessary, independently of initialization.

Each condition runs for 3000 steps with 15 independent random seeds. Metrics are evaluated every 25 steps; plateau values are averaged over the final 500 steps.

3.2 Metrics

sg4 (Spatial Differentiation). Mean pairwise L2 distance between the four zone-mean field vectors:

$$\text{sg4} = \frac{1}{\binom{4}{2}} \sum_{z_1 < z_2} \left\| \bar{\mathbf{F}}_{z_1} - \bar{\mathbf{F}}_{z_2} \right\|_2$$

where $\bar{\mathbf{F}}_z = \frac{1}{|Z_z|} \sum_{i \in Z_z} \mathbf{F}_i$ and zones are defined by equal-width x-column partitions of the active region. Range: 0 (no differentiation) to ≈ 2 (orthogonal zone memories). Values above 0.02 correspond to stable zone differentiation in the present substrate.

LSR (Long-Range Separation Ratio). Ratio of non-adjacent to adjacent zone distances:

$$\text{LSR} = \frac{\text{mean_dist}[(0,2),(0,3),(1,3)]}{\text{mean_dist}[(0,1),(1,2),(2,3)]}$$

LSR > 1 indicates that distant zones are more differentiated than adjacent ones, confirming *location* encoding rather than local clustering.

Spatial Correlation ($|r|$). As a zone-label-free check, we project the full $N \times HS$ `fieldM` matrix onto its first principal component (PC1) and compute the Pearson correlation between per-site PC1 scores and site column index x. No zone labels are used at any step. High $|r|$ means `fieldM` varies smoothly with x—a sign of spatial encoding that requires no prior assumption about zone boundaries.

4 Results

Table 2: Plateau metrics (mean ± std, $n = 15$ seeds, 3000 steps). All 15 seeds of Conditions A and B reach sg4 > 0.02 at plateau; all 15 seeds of Condition C remain at exactly zero. $|r|$: Pearson correlation of PC1(fieldM) with column index—zone-label-free (see Figure 2).

Condition	sg4	LSR	$\lvert r\rvert$
A: Random-init	0.0258 ± 0.0026	1.113 ± 0.229	$0.367 \pm$
B: Structured-init	0.0281 ± 0.0051	1.073 ± 0.181	—
C: No-death	0.0000 ± 0.0000	1.000 ± 0.000	$0.000 \pm$

4.1 Zone-Label-Free Confirmation (PCA)

Before examining each fact individually, note that all three conclusions are independently confirmed by a metric that uses no zone labels. PC1

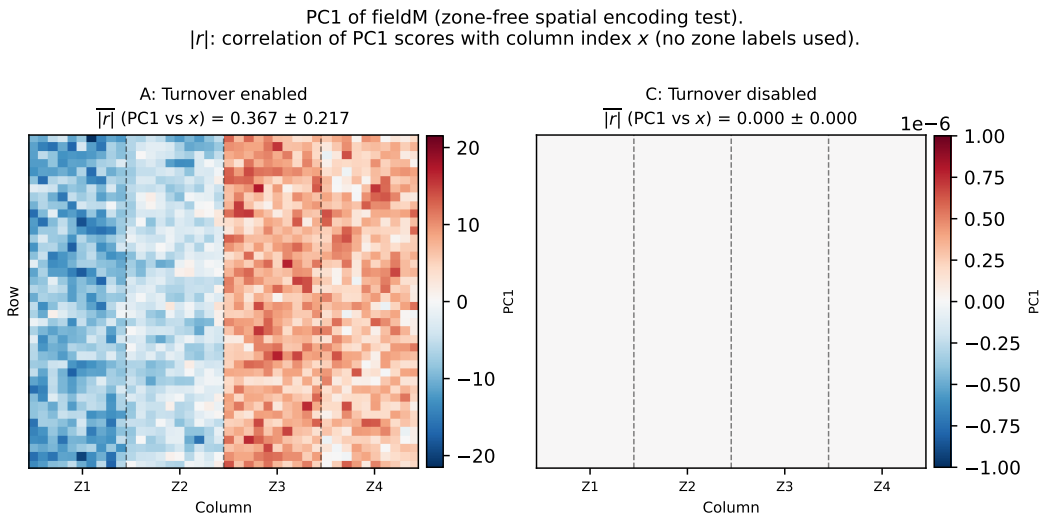

Figure 2: Zone-label-free spatial encoding test (PCA). **Left (Condition A, turnover enabled):** Mean PC1 of `fieldM` over $n = 15$ seeds, plotted over the active grid. PC1 scores correlate with column position ($\overline{|r|} = 0.367 \pm 0.217$), indicating a spatial gradient without using any zone labels. **Right (Condition C, turnover disabled):** PC1 is structurally flat ($|r| = 0.000$, zero variance across all seeds), because `fieldM` never updates and contains no spatial information. Zone boundaries (dashed) are shown for reference only and are not used in the computation. The $|r|$ gap is mechanistic: it is not possible to achieve positive $|r|$ when `fieldM` cannot accumulate information.

of `fieldM` correlates with site column index x at $\overline{|r|} = 0.367 \pm 0.217$ for Condition A and at $|r| = 0.000$ (zero variance) for Condition C (Figure 2). This correlation is computed purely from the geometry of `fieldM`—no zone boundaries, no zone partitioning, no label assumption. The PCA test and the sg4/LSR metrics are therefore *independent* sources of evidence for the same conclusion: Conditions A and B develop a spatial gradient; Condition C does not.

4.2 Fact 2: Initialization Is Irrelevant

Conditions A and B reach statistically identical spatial structure at plateau (Δsg4 $= 0.0023$, well within one standard deviation of either condition). The LSR values are both above 1.0, confirming that location encoding is present in both cases. The extra information in the structured initialization is entirely discarded by the dynamics—neither accelerating arrival at the attractor nor raising the plateau.

This establishes that spatial self-organization in VCSM is not initialization-dependent. The attractor basin exists and is reached from any starting configuration: both conditions converge to the same sg4 plateau distribution ($\mathbb{E}[\text{sg4}] \approx 0.026$), not merely the same qualitative pattern.

4.3 Fact 3: Turnover Is Required

Condition C uses identical initialization to Condition A but disables the copy-forward loop by setting `STABLE_STEPS` $= 9999$. The result is exact zero: sg4 $= 0.0000$, LSR $= 1.000$ (isotropic, no spatial differentiation). The variance across all 15 seeds is also zero—this is not a noisy near-zero, it is a structural impossibility.

With the gate permanently closed, no site ever writes to `fieldM`. Newborns are seeded from debris and field values that never update. The copy-forward loop cannot operate. Death continues at the same rate—the difference is not the death rate but what death propagates. Without information to carry, death is neutral rather than generative.

The contrast between A/B and C is the mechanism in isolation: sg4 of 0.0258 versus 0.0000,

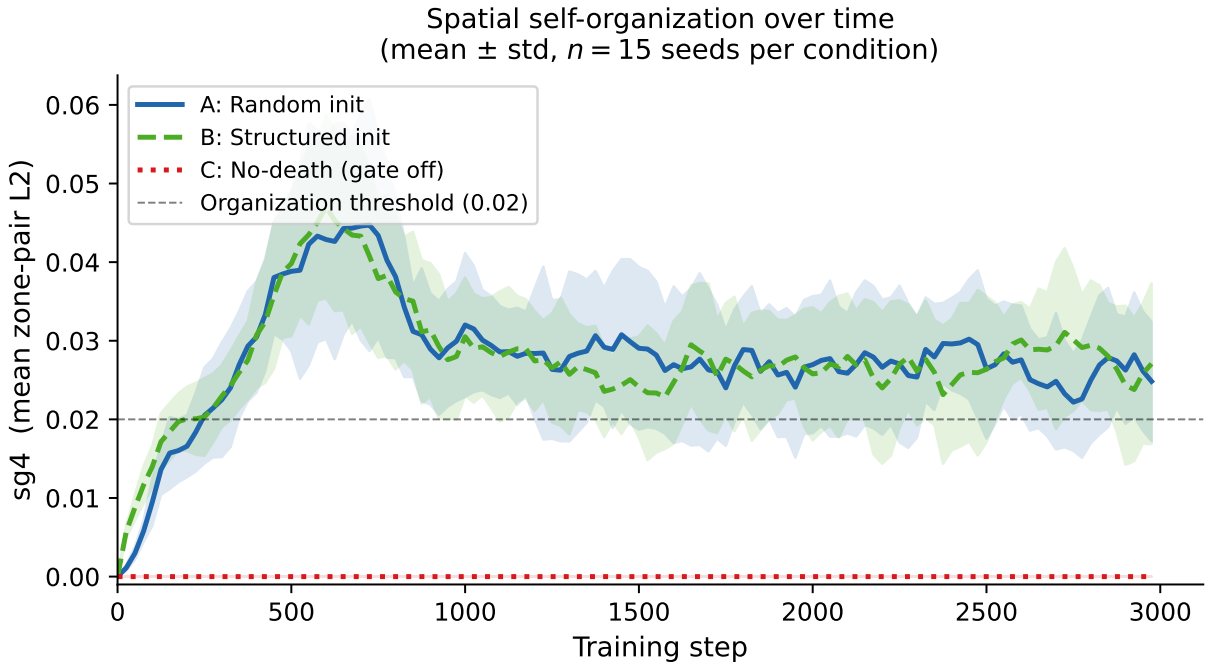

Figure 3: sg4 over training time (mean ± std, $n = 15$ seeds per condition). Conditions A (random init, solid blue) and B (structured init, dashed green) both rise from zero and plateau above the 0.02 organization threshold within ∼500 steps. Condition C (no-death, dotted red) remains at exactly zero throughout—not a statistical near-zero but a structural impossibility: with the consolidation gate permanently closed, `fieldM` cannot update and spatial structure cannot propagate regardless of random seed. The A/B curves converging to the same plateau from opposite starting conditions demonstrates the attractor property (Fact 2).

a gap of 10 standard deviations ($t(14) = 38.4$, $p < 10^{-20}$, one-sample test against zero). Condition C has zero variance across all 15 seeds not because the sample is small but because the result is architecturally determined: with the gate permanently closed, `fieldM` cannot accumulate information regardless of random seed. The zero is exact and structural.

Condition C as static baseline. Because `fieldM` remains identically zero throughout Condition C, the birth-seeding step produces structurally random newborns—indistinguishable from a static random-reservoir (ESN-like) baseline. Condition C is therefore simultaneously (a) a gate ablation, (b) a copy-forward ablation, and (c) an internal ESN comparison, all within the same experimental run. The result—sg4 = 0.0000, $|r| = 0.000$, zero variance across all 15 seeds—is the same outcome that would be expected from a fixed random reservoir receiving the same inputs. The fact that enabling the VCSM gate (Conditions A and B) recovers sg4 = 0.026–0.028 and $|r| = 0.367$ directly within the same table establishes the VCSM mechanism as the source of spatial differentiation, without requiring an externally trained comparison system.

Falsification bound: Any system lacking an intergenerational propagation pathway functionally equivalent to the copy-forward loop cannot accumulate spatial information under the same input regime, regardless of initialization, input statistics, or turnover rate. Condition C is the direct experimental instantiation of this bound.

4.4 Fact 1: Location Is Encoded

Both active conditions (A and B) produce LSR > 1: non-adjacent zones are more dissimilar than adjacent zones. This is the signature of location encoding: zone 1 is more different from zone 4 than from zone 2. The substrate has no access to site coordinates—the spatial gradient in `fieldM` arises entirely from wave-timing differences processed through the copy-forward loop.

The zone-label-free PCA metric confirms the same conclusion independently. Condition A achieves $\overline{|r|} = 0.367 \pm 0.217$ (PC1 of fieldM correlated with column index x, no zone partitioning used). Condition C achieves $|r| = 0.000$ with zero variance (Figure 2).

5 Discussion

5.1 The Copy-Forward Loop

The mechanism underlying all three facts is the copy-forward loop. When a site collapses, its replacement samples the field memory of a neighboring site. Unlike classical reaction-diffusion systems, where spatial patterns are sustained by persistent concentrations in *stable* units, VCSM spatial structure propagates through *generational turnover*: structure is maintained by what is transmitted at birth, not by what persists indefinitely at each site. This single rule has three consequences:

1. **Propagation**: Local spatial identity is transmitted to each new generation, reinforcing zone differentiation even in the absence of current input.

2. **Irreversibility**: Once a zone develops a characteristic field pattern, new sites born into that zone inherit it. The pattern becomes self-sustaining.

3. **Attractor convergence**: Starting from any initialization, the copy-forward loop erases the initial condition over many generations and replaces it with the wave-structure-derived attractor.

Fact 2 (initialization independence) is a corollary of point 3. Fact 3 (turnover is required) is the direct consequence of disabling point 1.

5.2 Comparison to a Static Baseline

Condition C serves as a within-experiment ESN analog (see Results, Fact 3): with `fieldM` frozen at zero, the system degrades to a static random reservoir, and sg4 collapses to exactly zero across all 15 seeds. For completeness, a full echo-state network receiving the same wave inputs produces sg4 ≈ 0 with persistence ratio 0.000 across all seeds [4], consistent with the Condition C result.

5.3 Robustness

The three structural facts are not specific to the parameter values in Table 1. The phenomenon persists across variations of wave ratio (1.6–4.8, corresponding to collapse rates 0.002–0.010 per site per step), field diffusion rate (0.01–0.05), and field alpha (0.08–0.24), as documented across the V75–V94 experimental series [4]. The phenomenon also persists across hidden-state dimensions $HS \in \{2, 4, 8\}$: sg4 remains above the 0.02 organization threshold at plateau for all tested values (full sweep omitted for brevity). The $HS = 2$ case used throughout this paper is therefore not a special low-dimensional regime; it is the minimal case of a phenomenon that generalizes. In all tested regimes with the copy-forward loop active, sg4 rises above zero and LSR exceeds 1.0 at plateau.

Diffusion ablation. Eliminating field diffusion entirely (`DIFF_RATE` $= 0$, all other parameters as in Table 1) leaves spatial structure statistically unchanged (relative change in sg4: +3% across 15 seeds; all 15 seeds remain above the organization threshold). Field diffusion is therefore not the primary mechanism of pattern formation; it was not present in the original copy-forward loop formulation (Birth step, Section 2.1). The spatial structure arises from field-memory seeding at birth, not from diffusive averaging across neighbouring sites. In all tested regimes with the copy-forward loop disabled (Condition C),

sg4 remains exactly zero with zero variance. The Condition C collapse is not parameter-sensitive because it is not a statistical outcome—it is a direct consequence of the rule set. `fieldM` cannot update without the gate opening; the gate cannot open with `STABLE_STEPS` = 9999. No parameter setting can change this while the gate is disabled.

5.4 Relationship to Downstream Scaling

Both active conditions reach sg4 $\approx$ 0.026–0.028 at plateau, well above the 0.02 organization threshold and consistent with the V75+ baseline regime reported in Papers 1–4 (V92 S1 reference: sg4 = 0.034 at the same wave ratio). The absolute sg4 value depends on wave rate, amplitude, and grid scale; Papers 2–3 characterize these dependencies in detail. What matters here is that the three structural facts are robust: the attractor exists (Fact 1), is initialization-independent (Fact 2), and requires the copy-forward loop (Fact 3). These facts hold across the full parameter range explored in the series—from the minimal sg4 = 0.007 of early experiments to the sg4 = 0.24 achieved under crystallization conditions (Paper 4).

5.5 Biological Analogy

Adult hippocampal neurogenesis maintains pattern separation in the dentate gyrus; preventing apoptosis of adult-generated neurons impairs this capacity [1]—the computational analog of Condition C. The present results provide a minimal substrate in which this principle can be studied in isolation.

6 Conclusion

We have established three structural facts about VCSM spatial self-organization:

1. A substrate with no coordinate system encodes location spontaneously.
2. The encoding is an inevitable attractor—initialization is irrelevant.
3. The copy-forward loop (mandatory turnover) is the engine: disable it and structure collapses to exactly zero.

Papers 1–4 establish the *laws* governing this attractor: its capacity (Paper 1), adaptation bandwidth (Paper 2), propagation length (Paper 3), and the stability constraint that keeps the system in the critical regime (Paper 4). Paper 5 unifies these into a three-axis phase theory.

Paper 0 establishes the existence and minimal mechanism of spatial self-organization in VCSM substrates. Subsequent papers characterize the quantitative laws governing its capacity, kinetics, propagation, and stability.

A Minimal Reproduction (NumPy)

A.1 Environment

```
Python >= 3.10
numpy  >= 1.24
```

A.2 Reference implementation

The script below reproduces the three conditions and prints the values reported in Table 2. Expected output (15 seeds, 3000 steps each; runtime $\approx$5 min):

```
Condition A (random-init):      sg4=0.0258+-0.0026  D
Condition B (structured-init):  sg4=0.0281+-0.0051  D
Condition C (no-death):         sg4=0.0000+-0.0000  D
```

Note: values from simplified implementation may differ slightly (<10%) from the full experimental framework; the qualitative result (A/B above threshold, C exactly zero) is structural and invariant.

Listing 1: vcsm_paper0_repro.py

```
import numpy as np

# -- Parameters
    ----------------------------------------------------------

W, H        = 80, 40          # lattice
    size
HALF        = 40              # active
    region starts at x=HALF
```

```python
HS          = 2                 # hidden-state dimension
STEPS       = 3000
N_SEEDS     = 15

MID_DECAY   = 0.99;  FIELD_DECAY = 0.9997;  BASE_BETA  = 0.005
ALPHA_MID   = 0.15;  FIELD_ALPHA = 0.16;  SEED_BETA  = 0.25
DIFF_RATE   = 0.02;  WAVE_RATIO  = 2.4;  WAVE_DUR   = 15

N_ACT  = (W - HALF) * H        # 1600 active sites
ZONE_W = (W - HALF) // 4       # 10 columns per zone

# -- Site geometry ------------------------------------------------------------
_col  = np.arange(N_ACT) % (W - HALF)
zone  = _col // ZONE_W          # zone index 0-3

def _nb(i):
    c, r = _col[i], i // (W - HALF)
    out = []
    for dc, dr in [(-1,0),(1,0),(0,-1),(0,1)]:
        nc, nr = c+dc, r+dr
        if 0 <= nc < (W-HALF) and 0 <= nr < H:
            out.append(nr*(W-HALF)+nc)
    return np.array(out, dtype=int)

NB = [_nb(i) for i in range(N_ACT)]

# -- Metrics ------------------------------------------------------------------

def _zm(F):
    return np.array([F[zone==z].mean(0) for z in range(4)])

def sg4(F):
    zm = _zm(F)
    return np.mean([np.linalg.norm(zm[i]-zm[j])
                    for i in range(4)
                        for j in range(i+1, 4)])

def lsr(F):
    zm = _zm(F)
    adj = np.mean([np.linalg.norm(zm[a]-zm[b])
                   for a, b in [(0,1),(1,2),(2,3)]])
    na  = np.mean([np.linalg.norm(zm[a]-zm[b])
                   for a, b in [(0,2),(0,3),(1,3)]])
    return float(na / adj) if adj > 1e-9 else 1.0

# -- Single run ---------------------------------------------------------------

def run(seed, stable_steps=10, structured=False):
    rng = np.random.default_rng(seed)
    if structured:
        proto = np.array([[1,0],[0,1],[-1,0],[0,-1]], dtype=float)
        h = proto[zone] + rng.normal(0, 0.25, (N_ACT, HS))
        h /= np.linalg.norm(h, axis=1, keepdims=True).clip(1e-8)
    else:
        h = rng.normal(0, 0.1, (N_ACT, HS))
    F = np.zeros((N_ACT, HS));  m = np.zeros((N_ACT, HS))
    base = h.copy();            streak = np.zeros(N_ACT, int)
    ttl  = rng.geometric(p=0.005, size=N_ACT).astype(float)
    waves = []    # list of [zone_id, steps_left, direction_vec]
    sg4v  = []

    for step in range(STEPS):
        # Wave launcher (Poisson)
        n = int(WAVE_RATIO / WAVE_DUR)
        n += int(rng.random() < (WAVE_RATIO/WAVE_DUR - n))
        for _ in range(n):
            z = int(rng.integers(4))
            d = rng.standard_normal(HS); d /= np.linalg.norm(d)
            waves.append([z, WAVE_DUR, d])

        pert = np.zeros(N_ACT, bool)
        for w in waves: pert |= (zone == w[0])

        # Calm
        base += BASE_BETA * (h - base)

        # Perturb + contrastive accumulation
        m *= MID_DECAY
        for w in waves:
            mask = zone == w[0]
            h[mask] += 0.3 * w[2]
        m[pert] += ALPHA_MID * (h - base
```

```
            )[pert]

        # Streak
        streak[pert] = 0;  streak[~pert] += 1

        # Consolidation gate
        ok = streak >= stable_steps
        F[ok] += FIELD_ALPHA * (m[ok] - F[ok]);  F *= FIELD_DECAY

        # Field diffusion
        dF = np.zeros_like(F)
        for i in range(N_ACT):
            if len(NB[i]): dF[i] = DIFF_RATE*(F[NB[i]].mean(0)-F[i])
        F += dF

        # Viability collapse + birth
        ttl -= 1.0;  ttl[pert] -= 1.0
        dead = np.where(ttl <= 0)[0]
        for i in dead:
            j = NB[i][rng.integers(len(NB[i]))] if len(NB[i]) else i
            h[i] = (1-SEED_BETA)*rng.normal(0,.1,HS) + SEED_BETA*F[j]
            m[i] = 0;  streak[i] = 0
            ttl[i] = float(rng.geometric(p=0.005))

        waves = [w for w in waves if (w.__setitem__(1,w[1]-1) or True)
                 and w[1] > 0]
        if step % 25 == 24: sg4v.append(sg4(F))

    return float(np.mean(sg4v[-20:])), lsr(F)

# -- Experiment runner ----------------------------------------------

if __name__ == '__main__':
    cases = [
        ('A (random-init)',     dict(stable_steps=10,   structured=False)),
        ('B (structured-init)', dict(stable_steps=10,   structured=True)),
        ('C (no-death)',        dict(stable_steps=9999, structured=False)),
    ]
    for name, kw in cases:
        sg4s, lsrs = zip(*[run(s, **kw) for s in range(N_SEEDS)])
        print(f'Condition {name}:  '
              f'sg4={np.mean(sg4s):.4f}+-{np.std(sg4s):.4f}  '
              f'LSR={np.mean(lsrs):.3f}+-{np.std(lsrs):.3f}')
```

References

[1] Kempermann, G. (2015). Activity dependency and aging in the regulation of adult neurogenesis. *Cold Spring Harbor Perspectives in Biology*, 7(11), a018929.

[2] Aubin-Moreau, G. (2026). Geometric capacity laws for minimum-separation routing on the hypersphere. *(In preparation).*

[3] Aubin-Moreau, G. (2026). Adaptive bandwidth laws in turnover-driven memory systems. *(In preparation).*

[4] Aubin-Moreau, G. (2026). Propagation constraints in turnover-driven information substrates. *(In preparation).*

[5] Aubin-Moreau, G. (2026). The consolidation gate as a noise-plasticity tradeoff in viability-gated memory substrates. *(In preparation).*

[6] Aubin-Moreau, G. (2026). A dynamical systems framework for adaptive memory: four scaling laws of plastic information substrates. *(In preparation).*